Proof-of-Concept Extraction of Many-Source Gravitational Readability Structure

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Abstract

The gravitational coherence surface (GCS) formalism defines a parity horizon of gravitational identity: the surface where a declared gravitational contribution reaches equality with its derived gravitational context under a declared readability rule. The many-source parity-network construction extends that logic by comparing weak-field source-readable contributions pairwise. A remaining question is operational: can derived readability structures be extracted from many-source weak-field configurations under an explicit comparison rule?

This paper reports three proof-of-concept extraction classes using the scalar tidal-readable branch $Q_i \propto M_i/r_i^3$. First, in an extreme-ratio configuration, a finite local readable pocket remains recoverable around a subordinate source inside a globally dominant environment. Second, dense many-source configurations can produce retained boundary-support architecture generated by source-readable competition rather than imposed tessellation. Third, matched-budget source ensembles can produce non-degenerate support responses under changes of spatial arrangement.

The results are interpreted conservatively. They do not imply modified gravity, a new force, a physical boundary, a new invariant, or direct observability. They establish a narrower threshold claim: under a declared weak-field readability rule, many-source gravitational scenes can yield operationally recoverable mesoscale boundary-support structure whose morphology responds to hierarchy, density, and source arrangement.

Keywords: general relativity, weak-field gravity, gravitational coherence surface, gravitational readability, parity extraction, many-source gravity, boundary-support geometry, mesoscale structure, morphology sensitivity

1 Introduction

The gravitational coherence surface program begins with a descriptive question: where does a declared gravitational contribution remain readable against the derived context of the scene in which it is embedded? The many-source parity construction extends that question from contribution-context parity to source-source parity. In that setting, pairwise equality surfaces and retained parity networks provide a mathematical language for source-readable balance.

Definitions alone, however, are not enough. A parity network that cannot be extracted is only a formal possibility. A proposed mesoscale readability layer that cannot be recovered, visualized, perturbed, or compared remains a mathematical sketch rather than a working diagnostic.

This paper therefore asks a narrower operational question:

Can derived readability structures be recovered from many-source weak-field configurations under a declared com

The answer reported here is yes, in a proof-of-concept sense. The examples are limited, and their interpretation is deliberately conservative. They establish operational recoverability under a declared branch, not branch-independent uniqueness, final invariance, dynamical authority, or observational status.

2 Scope and claim discipline

All results in this paper are interpreted within standard weak-field general relativity. No modification of gravity is proposed.

Let a finite source ensemble be declared on a weak-field time slice, with source masses M_i, positions x_i, and distances

$$r_i(x) = \|x - x_i\|.$$

The declared readable branch used in these proof-of-concept examples is the scalar tidal proxy

$$Q_i(x) \propto \frac{M_i}{r_i(x)^3}.$$

For sources i and j, the pairwise comparison field is

$$\Psi_{ij}(x) = Q_i(x) - Q_j(x),$$

and the corresponding parity surface is the zero set

$$\Sigma_{ij} = \{x : \Psi_{ij}(x) = 0\}.$$

Boundary-support samples are retained where the declared comparison rule identifies parity, low-margin structure, or related support behavior. Rendered point clouds or surfaces are downstream presentations of extracted supports. They are not the source of the result.

This paper does not claim that the reported structures are physical membranes, force barriers, geodesic tracks, new invariants, or direct observables. It also does not claim that every possible readable branch produces the same structure. The claim is operational: under the declared rule, structured support geometry can be recovered and compared across source configurations.

3 Minimal extraction frame

The proof-of-concept results use exact-source weak-field evaluations. Source contributions are computed from the declared source ensemble rather than from an imposed tessellation or coarse-grained density field. The readable contribution for source i follows the scalar branch $Q_i \propto M_i/r_i^3$.

The extraction logic is support-first. The primary objects are retained support samples and low-margin structures satisfying the declared comparison rule. Rendered point clouds or surfaces are visual presentations of those supports.

Three contexts are examined:

1. an extreme-ratio local-pocket test;
2. dense many-source boundary-support recovery;
3. matched-budget morphology comparison.

These contexts test hierarchy, emergence, and arrangement sensitivity.

4 Result 1: extreme-ratio local readable pocket

The first test examines whether a subordinate source can retain a local readable support region inside a globally dominant environment.

If extraction simply followed the largest contribution everywhere, the output would collapse into a trivial single-source wash. Instead, the recovered support shows a finite local pocket around the subordinate source, compressed and displaced by the dominant competitor.

The important point is not that the pocket is a physical wall. It is not. The point is that local readability is not erased merely because the broader scene is strongly hierarchical. The support remains finite, anisotropic, and shaped by the larger environment.

This establishes the first proof-of-concept criterion: persistence under hierarchy.

5 Result 2: dense many-source boundary-support architecture

The second test asks whether dense many-source competition yields extractable support architecture at all.

Under the declared readable comparison rule, retained support samples can assemble into interface-bearing morphology. The term *foam-like* may be useful as a visual shorthand, but it is not an ontological claim. No physical foam or microscopic material structure is implied. The recovered morphology indicates that boundary-support structure can emerge from source-readable competition rather than being imposed as a prior tessellation.

Because some early renders of dense many-source scenes were generated before later extraction and rendering corrections, this paper does not rely on those older images as evidence. The scientific claim is intentionally narrower than the visual metaphor: many-source support structure is operationally recoverable under the declared comparison rule, and its public presentation should remain downstream of validated extraction.

This establishes the second proof-of-concept criterion: emergence under many-source complexity.

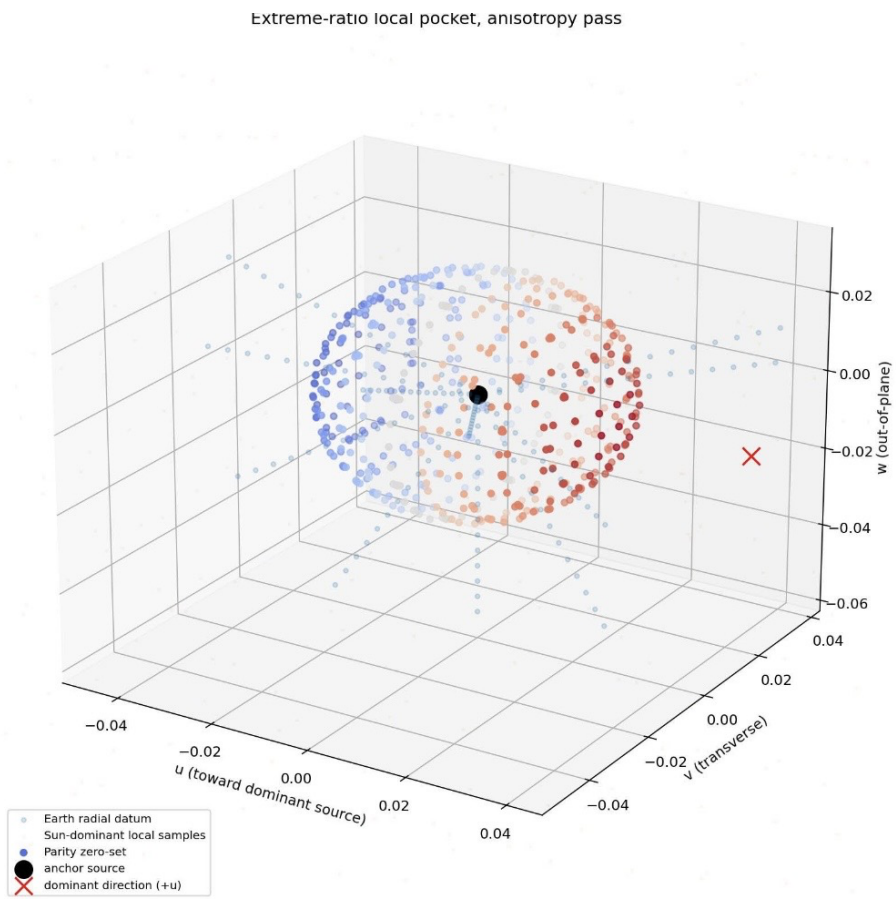

Figure 1: Extreme-ratio local readable pocket. A finite anisotropic support remains recoverable around a subordinate source in the presence of a dominant competitor. The support is diagnostic structure, not a force boundary. The plotted point cloud is a visualization of retained support samples under the declared readable branch.

6 Result 3: matched-budget morphology sensitivity

The third test asks whether recovered support geometry responds to source arrangement rather than only to total source budget.

Matched-budget source ensembles compare different spatial arrangements while holding the declared source budget fixed. In the working examples, smooth, weak-arm, and lopsided arrangements produce non-degenerate support responses. The smooth case gives a more even support response. The weak-arm case develops directional extension. The lopsided case shows stronger imbalance and preferential concentration.

This does not yet define a morphology invariant or a classification metric. It shows a more limited but important result: under matched-budget conditions, the support geometry can respond to arrangement rather than collapsing to the same bulk descriptor.

This establishes the third proof-of-concept criterion: sensitivity to morphology.

7 Cross-result synthesis

The three results form a minimal evidence triad.

The extreme-ratio pocket shows that local readable structure can persist under strong hierarchy. The dense many-source harvest shows that support geometry can emerge in complex source environments. The matched-budget comparison shows that recovered structure can respond to arrangement, not merely to total source count or total source budget.

Together, these results establish operational recoverability under the declared weak-field readability rule. That is the appropriate threshold claim for a first extraction paper.

Table 1: Proof-of-concept result triad. Each result tests a different failure mode of the mesoscale-readability hypothesis.

Result class	Diagnostic question	Threshold finding
Extreme ratio	Does hierarchy erase local readability?	Finite local support can persist.
Dense many-source	Can complex scenes yield support architecture?	Interface-bearing support can be recover
Matched budget	Does arrangement matter beyond total budget?	Responses can be non-degenerate.

8 Limits

The present results are preliminary.

They do not establish branch-independent uniqueness. A different readable operator may produce different support geometry.

They do not establish observational status. The extracted structures are derived diagnostics, not direct measurements.

They do not establish invariance. Stability under perturbation, sampling, domain, and branch changes remains to be tested systematically.

They do not establish a new physical object. The supports are induced structures inside a declared comparison rule.

They do not establish a final extraction standard. Certification criteria must be hardened in subsequent benchmark work.

9 Next phase

The next phase is benchmark hardening.

The equal-mass and unequal-mass two-source cases should be treated as analytic anchors. The equal-mass three-source case should serve as the first nontrivial many-source benchmark. Perturbed and unequal configurations should test continuous deformation. Dense configurations should be tested for persistence under refinement and source-roster changes.

The scalar branch $Q_i \propto M_i/r_i^3$ should remain the baseline until it is stable. Richer weak-field tidal tensor projections or norm-based branches can then be compared against it.

Only after stability is demonstrated should the results be correlated with downstream tasks such as trajectory comparison, residual analysis, or morphology-sensitive classification. Such correlations must be earned separately. They should not be assumed from the existence of recovered support geometry alone.

10 Non-claims

This paper does not propose modified gravity. It does not introduce a new force. It does not introduce physical membranes or walls. It does not claim that parity surfaces are geodesics. It does not claim that seams are attractors. It does not claim a new invariant. It does not replace numerical relativity. It does not replace averaging frameworks. It does not report observational detection.

It argues only that, under a declared weak-field readability rule, many-source scenes can contain recoverable derived support geometry worth stricter benchmark testing.

11 Conclusion

The GCS, many-source parity-network, and mesoscale-readability constructions define a diagnostic language for gravitational readability. This paper shows that the language can be made operational in first proof-of-concept form.

Under the declared weak-field readable rule $Q_i \propto M_i/r_i^3$, extraction recovers finite local support under hierarchy, interface-bearing support architecture under many-source competition, and non-degenerate response under matched-budget morphology variation.

These results do not change gravity. They do not define a new observable or force. They show something narrower and useful: many-source weak-field scenes can contain recoverable derived support geometry that deserves stricter benchmark testing.

The next task is not expansion. It is hardening.

A Minimal conventions

All examples use standard weak-field source evaluation under the declared scalar readable branch

$$Q_i \propto \frac{M_i}{r_i^3}.$$

Pairwise comparison is represented by

$$\Psi_{ij} = Q_i - Q_j.$$

Retained supports are downstream of the declared comparison rule. Rendered figures should be read as visualizations of extracted supports, not as the extraction itself.

B Figure note

Only the extreme-ratio local-pocket figure is included in this public release. Earlier dense-scene and matched-budget graphics are treated as superseded presentation artifacts until regenerated under the corrected rendering path. This choice is conservative: the paper preserves the operational threshold claims while avoiding visual overcommitment.

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